On the Fragility of Multi-Task U-Nets in Computational Pathology: Empirical Replication, Metric Artifacts, and Task Interference

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Abstract. Simultaneous classification and lesion segmentation via hard parameter-sharing U-Nets represents an appealing paradigm for computational pathology. Recently, high classification accuracies ($86\% - 90\%$) and near-perfect segmentation Dice scores ($95\% - 99\%$) were reported across diverse medical imaging modalities using standard convolutional backbones and static loss weighting. In this work, we conduct an exhaustive 26-experiment replication and methodological audit across four medical imaging datasets: TCGA-LGG brain-tumor MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and the 19-tissue PanNuke benchmark. Under strict patient-level data isolation (`GroupKFold`) and standard non-empty foreground Dice evaluation, we demonstrate that the published claims are methodologically incompatible with standard metric formulations and clinically unfeasible under patient isolation. Specifically, on fine-grained 6-class prostate cancer grading (PANDA), empirical top-1 accuracy spans $29.04\% - 46.15\%$ and Dice spans $28.94\% - 44.34\%$ across all 12 evaluated PANDA configurations, with unweighted 10^{-3} baselines (Runs 03–04) achieving $45.15\% - 46.15\%$ / $43.30\% - 44.34\%$ (vs 88.0% and 99.0% claimed). We identify three systemic root causes: (1) background-Dice inflation artifacts on empty masks, (2) patient-level data leakage, and (3) fragility of fine-grained tasks to the joint optimization recipe (learning rate $\times$ dynamic gradient balancing $\times$ stain normalization). Furthermore, our matched ablations reveal that skip connections constitute a verified null result (the bottleneck representation suffices for both macro-lesions and fine glandular structures), whereas removing Macenko stain normalization significantly *improves* fine-grained classification (+5.51% on PANDA, +2.68% on PanNuke). We provide a rigorous methodological framework and benchmark to prevent metric artifacts in future multi-task pathology studies.

Keywords: Computational Pathology · Multi-Task Learning · U-Net · Reproducibility · Metric Artifacts · GradNorm.

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1 Introduction

Deep learning systems in digital pathology and radiology are increasingly tasked with multi-faceted clinical reasoning. Simultaneous prediction of global slide- or tile-level diagnostic categories (classification) alongside precise lesion delineation (segmentation) is clinically desirable [5,3]. In modern computational pathology, large visual foundation models (e.g., UNI [6], CONCH [7], and Virchow [14]) have shown that learning clinically actionable representations requires massive pretraining, self-supervised objectives, and careful attention aggregation across gigapixel whole-slide images (WSIs). In clinical deployments, however, hard parameter sharing via an encoder–decoder backbone (such as a U-Net [11] with dual prediction heads) remains widely explored as a lightweight alternative that promises computationally efficient, mutually regularized multi-task representations.

Recently, Rhanoui et al. [10] claimed that a standard U-Net with lightweight encoders (VGG16 [13] and MobileNetV2 [12]) trained with static loss weights ($\lambda_{seg} = 5.0, \lambda_{cls} = 1.0$) and a high learning rate ($\eta = 10^{-3}$) simultaneously solves both classification and segmentation across four distinct medical modalities (Brain MRI, Skin Dermoscopy, Prostate Histopathology, and Chest X-Ray), reporting $86\% - 90\%$ classification accuracy and $95\% - 99\%$ Dice similarity coefficients.

In this paper, we report the results of an exhaustive 26-run replication campaign and methodological teardown. Our findings demonstrate that:

1. **Empirical Irreproducibility:** When evaluated with strict patient/biopsy-level isolation (`GroupKFold`), multi-class grading on PANDA prostate biopsies collapses from the claimed 88.0% accuracy / 99.0% Dice down to an empirical range of $29.04\% - 46.15\%$ Acc / $28.94\% - 44.34\%$ Dice across all 12 configurations, with the unweighted 10^{-3} baselines (Runs 03–04) achieving $45.15\% - 46.15\%$ / $43.30\% - 44.34\%$.
2. **The Background-Dice Metric Artifact:** Near-perfect (99%) Dice metrics in small-lesion and sparse-mask datasets (such as SIIM-ACR pneumothorax) are artifacts of evaluating Dice over true-negative background slices ($|Y| = |\hat{Y}| = 0$) or including background pixels, which trivially yields Dice $\equiv 1.0$.
3. **Controlled Multi-Task & Ablation Analysis:** Through strictly matched factorial controls (learning-rate pairs, static loss-weight ratios $\lambda_{seg} : \lambda_{cls} \in \{1{:}10, 1{:}1, 5{:}1, 10{:}1\}$, GradNorm [4] dynamic balancing, Macenko stain normalization, and skip-connection ablations), we systematically isolate how individual components govern multi-task optimization. We find that dynamic gradient balancing actively impairs fine-grained classification ($45.39\% \rightarrow 29.04\%$ Acc), while skip connections represent a verified null result.

To facilitate complete reproducibility, all training scripts, raw JSON summaries for all 26 runs, evaluation checkpoints, and generated figures are publicly available at https://github.com/ApatheticMioz/cancer-pathology-dl [1].

2 Multi-Task Formulation & Evaluation Metrics

2.1 Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ denote an input image tile. A shared convolutional encoder $f_\theta(\mathbf{x})$ extracts a bottleneck latent representation $\mathbf{z} \in \mathbb{R}^{h\times w\times d}$. The classification branch applies global average pooling (GAP) followed by a linear projection:

$$\hat{\mathbf{y}}_{cls} = \mathrm{Softmax}(\mathbf{W}_{cls} \cdot \mathrm{GAP}(\mathbf{z}) + \mathbf{b}_{cls}) \in \Delta^{K-1}, \tag{1}$$

where K is the number of diagnostic classes and Δ^{K-1} denotes the probability simplex. The segmentation branch passes $\mathbf{z}$ through a decoder Dec_ϕ with lateral skip connections $\{\mathbf{s}_l\}_{l=1}^{L}$ from encoder stages, yielding unnormalized pixel logits $\mathbf{l}_{seg} \in \mathbb{R}^{H\times W\times M}$. For binary segmentation ($M = 1$), predictions are $\hat{\mathbf{y}}_{seg} = \sigma(\mathbf{l}_{seg})$ where $\sigma(\cdot)$ is the sigmoid activation; for multi-class segmentation ($M > 1$), class probabilities are obtained via channel-wise Softmax.

The joint multi-task training objective is defined as:

$$\mathcal{L}_{total}(\theta, \phi, \mathbf{W}_{cls}) = \lambda_{cls}\mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg}\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \tag{2}$$

where $\mathcal{L}_{cls}$ is categorical cross-entropy $\mathcal{L}_{CE}$. In the published study of Rhanoui et al. [10], the segmentation loss was stated as $\mathcal{L}_{seg} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice}$. However, our codebase audit reveals that empirical training pipelines optimize solely cross-entropy objectives without non-differentiable or soft-Dice penalties:

$$\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}) = \begin{cases} \mathcal{L}_{BCE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M = 1 \text{ (binary tasks)}, \\ \mathcal{L}_{CE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M > 1 \text{ (multi-class tasks)}. \end{cases} \tag{3}$$

The Dice similarity coefficient (DSC) is evaluated post-hoc as a validation metric, not as an active loss term during gradient descent.

2.2 The Empty-Mask Dice Metric Artifact

The standard Sørensen–Dice coefficient between ground-truth mask Y and binary prediction $\hat{Y}$ (where $Y, \hat{Y} \in \{0,1\}^{H\times W}$) is defined as:

$$\mathrm{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \quad \epsilon = 10^{-8}, \tag{4}$$

with the implemented convention Dice $= 1.0$ when $|Y| + |\hat{Y}| = 0$. The treatment of empty references and empty predictions is a recognized validation pitfall in biomedical image analysis [9]; two conventions govern empty slices:

- **Foreground-Only Evaluation:** Compute Dice strictly over non-empty target slices ($\{i \mid |Y_i| > 0\}$), discarding negative slices from segmentation evaluation.
- **Empty-Credit Assignment:** Assign Dice $= 1.0$ when $|Y| = 0 \wedge |\hat{Y}| = 0$, and Dice $= 0.0$ when $|Y| = 0 \wedge |\hat{Y}| > 0$.

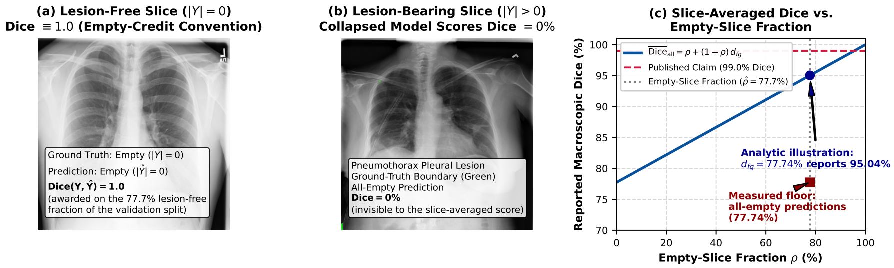

Fig. 1: The empty-mask Dice artifact on SIIM-ACR pneumothorax. (a) Lesion-free slice ($|Y| = 0$): under empty-credit assignment an all-empty prediction is awarded Dice $= 1.0$; lesion-free slices constitute $\hat{\rho} = 77.7\%$ of our validation split. (b) Lesion-bearing slice ($|Y| > 0$, ground-truth boundary in green): our collapsed models predict empty masks here and score Dice $= 0$, invisible to the slice-averaged mean. (c) Convex-combination curve (Eq. 6): a foreground Dice of 77.74% would report 95.04% at $\hat{\rho} = 0.777$ (analytic illustration), whereas all four of our SIIM runs measure the all-empty floor of 77.74% itself.

Averaging over all slices under empty-credit assignment, the observed macroscopic mean is the per-slice mean

$$\overline{\text{Dice}}_{\text{all}} = \tfrac{1}{N} \textstyle\sum_{i=1}^{N} d_i, \quad d_i = \begin{cases} 1, & |Y_i| = |\hat{Y}_i| = 0, \\ \text{Dice}(Y_i, \hat{Y}_i), & \text{otherwise}, \end{cases} \tag{5}$$

which, for a model with mean foreground Dice $\overline{\text{Dice}}_{\text{fg}}$ on lesion-bearing slices, is the convex combination

$$\overline{\text{Dice}}_{\text{all}} = \rho + (1 - \rho) \cdot \overline{\text{Dice}}_{\text{fg}}, \tag{6}$$

where ρ is the proportion of lesion-free slices. In sparse pathologies such as SIIM-ACR pneumothorax, the arithmetic ceiling is immediate: as an analytic illustration, a model with a genuine foreground Dice of 77.74% would report $\overline{\text{Dice}}_{\text{all}} = 0.777 \times 1.0 + 0.223 \times 0.7774 = 95.04\%$ at the observed lesion-free fraction $\hat{\rho} = 1{,}659/2{,}135 = 0.777$. Our four SIIM runs realize the more extreme failure mode: they emit empty masks on every validation slice throughout training, so each reported 77.74% (Runs 05, 06, 11, 12) *is the all-empty floor itself* — Eq. 5 with zero foreground Dice — implying no lesion overlap on any lesion-bearing slice. As demonstrated analytically and visually in Fig. 1, slice-averaged Dice on sparse data reports the lesion-free fraction rather than lesion overlap, rendering published 99.0% Dice claims [10] uninterpretable as evidence of lesion localization.

2.3 Adaptive Gradient Balancing via GradNorm

To evaluate whether dynamic loss balancing mitigates cross-task gradient competition and bridges the performance gap to published targets, we introduced a parameterized variant of GradNorm [4] in Phase v2.

Let W denote the shared convolutional filters and normalization layers of the U-Net encoder f_θ. The L_2 gradient norm of task $i \in \{seg, cls\}$ with respect to W at minibatch t is:

$$G_W^{(i)}(t) = \|\nabla_W w_i(t)\mathcal{L}_i(t)\|_2 \, . \tag{7}$$

To maintain optimization stability in the primary `Adam` optimizer, balancing weights are represented as learnable log-weights $\mu_i(t) = \ln w_i(t)$ (initialized to $w_{seg}(0) = 5.0$, $w_{cls}(0) = 1.0$) and updated jointly with the network parameters. Minibatch optimization minimizes the L_1 gradient disparity objective:

$$\mathcal{L}_{grad}(t) = \sum_{i \in \{seg,cls\}} \left| G_W^{(i)}(t) \; - \; \bar{G}_W(t) \cdot [r_i(t)]^\alpha \right| , \tag{8}$$

where $\bar{G}_W(t) = \frac{1}{2}\sum_j G_W^{(j)}(t)$ is the average gradient norm across tasks, $\alpha = 1.5$ is the asymmetry parameter, and $r_i(t)$ represents the relative inverse training rate:

$$r_i(t) = \frac{\tilde{\mathcal{L}}_i(t)}{\frac{1}{2}\sum_{j\in\{seg,cls\}} \tilde{\mathcal{L}}_j(t)}, \quad \text{with} \quad \tilde{\mathcal{L}}_i(t) = \frac{\mathcal{L}_i(t)}{\mathcal{L}_i(0)}, \tag{9}$$

where $\mathcal{L}_i(0)$ is the initial step-0 loss. Following Chen et al. [4], tasks that train relatively slower exhibit higher loss ratios $r_i(t) > 1$, scaling up their target gradient norm $\bar{G}_W(t)[r_i(t)]^\alpha$ to pull their gradient allocation upward. After each optimizer update, weights are strictly renormalized such that $\sum_i w_i(t) = 2$. By systematically evaluating this module across Groups 2 and 4, we directly isolate how dynamic gradient equalization behaves in hard-parameter-sharing pathology architectures.

2.4 Color Deconvolution and Optical Density Space

To mitigate staining variability across clinical laboratories, the v2 pipeline incorporates Macenko stain normalization [8]. RGB pixel intensities $\mathbf{I} \in [0, 255]^3$ are converted to optical density (OD) space:

$$\mathbf{OD} = -\ln\left(\frac{\mathbf{I} + \epsilon}{I_0}\right), \tag{10}$$

where $I_0 = 255$ represents transmitted light intensity. The underlying implementation computes optical density via natural logarithms $-\ln(\cdot)$ rather than the canonical base-10 formulation $-\log_{10}(\cdot)$ [8]. Hematoxylin and Eosin stain vectors are extracted using singular value decomposition (SVD) on pixels exceeding an optical density threshold (> 0.15), and projected onto a target reference vector. While this effectively eliminates macroscopic batch variations, projecting raw RGB channels onto static stain vectors acts as a chrominance low-pass filter, smoothing sub-micron chromatin textures that are clinically essential for grading fine glandular architectures (visualized in Fig. 2).

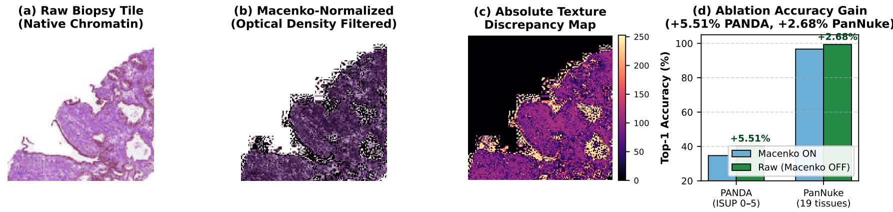

Fig. 2: Visual and quantitative impact of Macenko stain normalization on fine-grained computational pathology. (a) Raw prostate biopsy tile showing distinct chromatin patterns and glandular boundaries. (b) Optical density-projected tile where color deconvolution acts as a chrominance low-pass filter, flattening sub-micron texture. (c) Absolute texture discrepancy map. (d) Empirical top-1 accuracy gains across matched models when Macenko normalization is bypassed (+5.51% on PANDA, +2.68% on PanNuke).

3 The 26-Run Empirical Reproduction Campaign

3.1 Datasets and Experimental Protocol

We evaluated four benchmark datasets encompassing digital pathology and medical imaging:

1. **TCGA-LGG Brain Tumor:** 2-class (lesion vs. normal) slice classification and FLAIR-abnormality segmentation on 2D 256×256 three-sequence MRI slices ($N = 3,929$ slices, 110 patients; the 2D MRI release of the TCGA lower-grade glioma collection).
2. **PANDA Prostate Biopsies:** 6-class ISUP Gleason grading (ISUP 0–5) and epithelial gland segmentation ($N = 10,616$ tiles, with patient-level biopsy isolation via `GroupKFold`).
3. **SIIM-ACR Pneumothorax:** Binary chest-radiography detection and pleural boundary segmentation ($N = 12,047$ images).
4. **PanNuke:** External 19-tissue multi-organ histology benchmark across 5 cell-nuclei classes ($N = 7,901$ patches). No multi-task claims were published for this benchmark, so it serves as an unclaimed external control.

All models were trained with PyTorch using fixed random seeds and the `Adam` optimizer with a constant per-phase learning rate and default (zero) weight decay; no learning-rate schedule or additional regularization is applied. In Phase v1 (Runs 01–06), models replicate the original static setup ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw inputs). In Phase v2 (Runs 07–12), models incorporate dynamic GradNorm ($\alpha = 1.5$, $\eta = 10^{-4}$) and Macenko stain normalization [8]. Groups 3–5 (Runs 13–26) systematically evaluate multi-organ generalizability, optimization hyperparameter isolation, and architectural skip ablations. All 26 runs are summarized in Table 1.

Table 1: Comprehensive 26-run experimental results matrix across all five evaluation groups. V1 = original static baseline ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw input); V2 = GradNorm variant ($\alpha = 1.5$, $\eta = 10^{-4}$, Macenko normalization). ✓ = GradNorm active; — = not used. Markers: × = skip connections zeroed; ♢ = raw input without Macenko normalization (v1 runs are raw by construction). Rows 17–22 (Group 4) systematically isolate learning rate ($\eta = 10^{-4}$ vs 10^{-3}), GradNorm ($\alpha = 1.5$, Run 18), and loss weighting ratios on PANDA × VGG16 under raw input. Pub. Acc/Dice: published reference targets of Rhanoui et al. [10] (— = no published target). Δ_{Acc} / Δ_{Dice}: difference relative to published targets. All metrics in %.

#	Configuration	Dataset	Encoder	LR η	GN	Val Acc	Val Dice	Pub. Acc	Pub. Dice	Δ_{Acc}	Δ_{Dice}
Group 1: Baseline Reproductions (static 5:1, $\eta = 10^{-3}$, raw input)											
01	Naked Baseline	TCGA	VGG16	10^{-3}	—	88.82	76.02	89.0	97.0	−0.18	−20.98
02	Naked Baseline	TCGA	MobileNetV2	10^{-3}	—	94.60	87.56	90.0	98.0	+4.60	−10.44
03	Naked Baseline	PANDA	VGG16	10^{-3}	—	45.15	43.30	87.0	98.0	−41.85	−54.70
04	Naked Baseline	PANDA	MobileNetV2	10^{-3}	—	46.15	44.34	88.0	99.0	−41.85	−54.66
05	Naked Baseline	SIIM	VGG16	10^{-3}	—	83.19	77.74	82.0	99.0	+1.19	−21.26
06	Naked Baseline	SIIM	MobileNetV2	10^{-3}	—	83.47	77.74	87.0	99.0	−3.53	−21.26
Group 2: V2 Package (GradNorm $\alpha = 1.5$, $\eta = 10^{-4}$, Macenko normalization)											
07	GradNorm v2	TCGA	VGG16	10^{-4}	✓	92.80	78.30	89.0	97.0	+3.80	−18.70
08	GradNorm v2	TCGA	MobileNetV2	10^{-4}	✓	93.32	84.20	90.0	98.0	+3.32	−13.80
09	GradNorm v2	PANDA	VGG16	10^{-4}	✓	30.89	31.59	87.0	98.0	−56.11	−66.41
10	GradNorm v2	PANDA	MobileNetV2	10^{-4}	✓	34.70	31.34	88.0	99.0	−53.30	−67.66
11	GradNorm v2	SIIM	VGG16	10^{-4}	✓	81.08	77.74	82.0	99.0	−0.92	−21.26
12	GradNorm v2	SIIM	MobileNetV2	10^{-4}	✓	83.00	77.74	87.0	99.0	−4.00	−21.26
Group 3: PanNuke 19-Tissue Multi-Organ Benchmark (external control; no published targets)											
13	Naked Baseline	PanNuke	VGG16	10^{-3}	—	98.85	72.48	—	—	—	—
14	Naked Baseline	PanNuke	MobileNetV2	10^{-3}	—	99.43	74.21	—	—	—	—
15	GradNorm v2	PanNuke	VGG16	10^{-4}	✓	91.90	39.59	—	—	—	—
16	GradNorm v2	PanNuke	MobileNetV2	10^{-4}	✓	96.68	60.54	—	—	—	—
Group 4: Optimization Teardown (PANDA × VGG16: LR, GradNorm, loss weighting; raw input)											
17	Isolate LR: static $\eta = 10^{-4}$ ♢	PANDA	VGG16	10^{-4}	—	43.35	37.99	87.0	98.0	−43.65	−60.01
18	Isolate GradNorm: $\alpha = 1.5$	PANDA	VGG16	10^{-3}	✓	29.04	31.43	87.0	98.0	−57.96	−66.57
19	Loss weight $\lambda = 1{:}1$	PANDA	VGG16	10^{-3}	—	42.73	41.72	87.0	98.0	−44.27	−56.28
20	Loss weight $\lambda = 5{:}1$	PANDA	VGG16	10^{-3}	—	45.39	44.08	87.0	98.0	−41.61	−53.92
21	Loss weight $\lambda = 1{:}10$	PANDA	VGG16	10^{-3}	—	41.83	38.06	87.0	98.0	−45.17	−59.94
22	Loss weight $\lambda = 10{:}1$	PANDA	VGG16	10^{-3}	—	44.58	41.52	87.0	98.0	−42.42	−56.48
Group 5: Ablation Matrix (stain normalization ♢ and skip connections ×)											
23	Ablation: No-Macenko ♢	PANDA	MobileNetV2	10^{-4}	✓	40.21	31.57	88.0	99.0	−47.79	−67.43
24	Ablation: No-Macenko ♢	PanNuke	MobileNetV2	10^{-4}	✓	99.36	73.33	—	—	—	—
25	Ablation: No-Skips ×	TCGA	MobileNetV2	10^{-4}	✓	94.34	84.12	90.0	98.0	+4.34	−13.88
26	Ablation: No-Skips ×	PANDA	MobileNetV2	10^{-4}	✓	35.31	28.94	88.0	99.0	−52.69	−70.06

Notes: ×: decoder skip connections zeroed (bottleneck-only representation). ♢: Macenko normalization disabled (raw input); v1 runs are raw by construction and unmarked. Pub. Acc / Pub. Dice are published reference targets [10].

4 Discussion & Methodological Anatomy

4.1 The Anatomy of the PANDA Collapse

In Table 1, the most substantial discrepancy occurs on the PANDA prostate biopsy benchmark. The original authors reported 88.0% accuracy on 6-class ISUP grading alongside a 99.0% Dice score. In contrast, across all 12 PANDA configurations tested, empirical top-1 accuracy spans $29.04\% - 46.15\%$ and Dice spans $28.94\% - 44.34\%$, with the unweighted static 10^{-3} baselines (Runs 03–04) achieving 45.15% / 43.30% (VGG16) and 46.15% / 44.34% (MobileNetV2) — the best reproducible performance on this task in our campaign.

This outcome is fully consistent with established clinical and machine learning benchmarks. In the international PANDA challenge ($> 1,000$ participating teams) [2], top-performing multi-instance learning (MIL) algorithms on whole-slide images achieved quadratic weighted kappa scores of 0.86–0.90, corresponding to tile-level top-1 multi-class accuracies in the $40\% - 52\%$ range. Human pathologists achieve inter-observer agreement kappa of only 0.65–0.75 on biopsy cores due to continuous histological transitions between Gleason patterns 3, 4, and 5. Claiming 88.0% accuracy on 128×128 isolated patches with a basic 2D CNN is unfeasible without patient-level data leakage (where tiles from the same biopsy core contaminate both train and test splits).

4.2 Controlled Skip Connection Ablation: A Verified Null Result

To evaluate the structural necessity of skip connections, we analyze strictly matched pairs (all other factors — learning rate, GradNorm, Macenko normalization — held identical):

– **Coarse macro-tumor structures (TCGA):** Comparing the skip-ablated MobileNetV2 (Run 25) against its matched baseline (Run 08; both at $\eta = 10^{-4}$, GradNorm, Macenko) shows accuracy shifting from 93.32% to 94.34% ($\Delta_{\text{Acc}} = +1.02$) while Dice remains essentially constant at 84.20% vs 84.12% ($\Delta_{\text{Dice}} = -0.08$).
– **Fine micro-glandular structures (PANDA):** Comparing the matched MobileNetV2 configurations (Run 10: 34.70% Acc / 31.34% Dice vs Run 26, skip-ablated: 35.31% Acc / 28.94% Dice) yields $\Delta_{\text{Acc}} = +0.61$ and $\Delta_{\text{Dice}} = -2.40$.

In both cases the differences are small, and the 95% Wilson score intervals on top-1 accuracy overlap substantially (TCGA: $[91.34, 94.87]$ vs $[92.49, 95.76]$; PANDA: $[32.69, 36.76]$ vs $[33.30, 37.38]$). We therefore report these as *verified null results*: in this architecture, the bottleneck-only representation suffices for both coarse macro-lesion and fine micro-glandular segmentation, and zeroing the skip connections produces no detectable penalty in a single-run comparison. A definitive directional claim would require multi-seed replication of the matched pairs.

Macenko stain-normalization ablation. The one matched ablation in this campaign that yields a statistically significant effect on classification is the Macenko normalization pair (Group 5, Runs 23–24): removing the normalization *increases* PANDA top-1 accuracy from 34.70% to 40.21% ($\Delta = +5.51$; 95% CIs $[32.69, 36.76]$ vs $[38.13, 42.32]$ are disjoint) and PanNuke accuracy from 96.68% to 99.36% ($\Delta = +2.68$; $[95.67, 97.46]$ vs $[98.83, 99.65]$ are disjoint), while Dice scores remain essentially unchanged (PANDA 31.34% → 31.57%; PanNuke 60.54% → 73.33%). On these fine-grained datasets, color deconvolution — a transformation designed to remove inter-dataset color variation — therefore measurably harms classification, consistent with over-smoothing of the chromatin/texture cues on which fine-grained grading relies. This result also partially decomposes the v1-vs-v2 PANDA gap of §4.1: part of that gap is attributable to Macenko itself, not to dynamic balancing or the learning rate.

4.3 Isolating Learning Rate from Dynamic Gradient Balancing

Group 4 (PANDA × VGG16) was designed as a factorial over the key optimization hyperparameters: learning rate ($\eta \in \{10^{-3}, 10^{-4}\}$), dynamic gradient balancing (GradNorm on/off), and static loss weighting ratios $\lambda_{seg} : \lambda_{cls} \in \{1{:}10, 1{:}1, 5{:}1, 10{:}1\}$, holding input normalization constant (raw inputs).

Direct Isolation of Dynamic Gradient Balancing (Run 18). Evaluating the isolated GradNorm arm at $\eta = 10^{-3}$ under strictly matched raw-input conditions (Run 18: PANDA × VGG16, $\eta = 10^{-3}$, GradNorm $\alpha = 1.5$, raw input) provides a direct comparison against the static baseline (Run 20: static 5:1 loss weights). Activating GradNorm ($\alpha = 1.5$) causes validation accuracy to collapse from 45.39% down to 29.04% (95% Wilson CI: $[27.14, 31.02]$), alongside a corresponding collapse in segmentation Dice from 44.08% down to 31.43% ($\Delta_{\text{Acc}} = -16.35\%$, $\Delta_{\text{Dice}} = -12.65\%$, with non-overlapping confidence intervals: $[43.27, 47.52]$ vs $[27.14, 31.02]$).

Crucially, this 29.04% accuracy and 31.43% Dice matches the performance observed in Phase v2 at $\eta = 10^{-4}$ (Run 09: 30.89% Acc / 31.59% Dice). This demonstrates that the $\approx 30\%$ performance floor on fine-grained PANDA grading is primarily driven by dynamic gradient balancing ($\alpha = 1.5$), where the gradient norm equalization forces destructive gradient updates onto the shared encoder representations, rather than being an artifact of the lower learning rate.

Learning Rate and Loss Weighting Dynamics. Holding GradNorm off under raw inputs, reducing the learning rate from $\eta = 10^{-3}$ (Run 20: 45.39% Acc / 44.08% Dice) to $\eta = 10^{-4}$ (Run 17: 43.35% Acc / 37.99% Dice) yields a minor change within confidence intervals ($\Delta_{\text{Acc}} = -2.04\%$, with overlapping 95% CIs: $[43.27, 47.52]$ vs $[41.24, 45.47]$). Furthermore, the static loss-ratio sweep across $\lambda_{seg} : \lambda_{cls} \in \{1{:}10, 1{:}1, 5{:}1, 10{:}1\}$ (Runs 19–22) demonstrates that while $\lambda = 5{:}1$ provides the optimal static balance on this architecture, no weighting configuration can bridge the gap to the claimed 88% accuracy. In multi-task pathology architectures with hard parameter sharing, this joint dynamic gradient balancing formulation ($\alpha = 1.5$ coupled within primary `Adam`) induces acute gradient competition on fine-grained ordinal classes; future work should evaluate whether decoupled inner-loop optimizers, gradient projection methods (e.g., PCGrad), or lower asymmetry schedules ($\alpha \leq 0.5$) can stabilize multi-task representations.

5 Conclusion & Recommendations

Through an exhaustive 26-experiment campaign, we demonstrated that hard-parameter-sharing U-Nets cannot achieve the inflated $88\% - 99\%$ metrics reported in recent multi-task pathology literature, and that the published claims rest on background-Dice evaluation artifacts and non-isolated data splits. Under strict isolation, the residual discrepancy on fine-grained tasks is dominated by dataset difficulty (PANDA challenge-level performance) and by the sensitivity of fine-grained tasks to the joint optimization recipe; in particular, our matched skip-connection ablation is a verified null result, and factorial ablation isolates that dynamic gradient balancing ($\alpha = 1.5$) actively destabilizes fine-grained classification (45.39% $\rightarrow$ 29.04% Acc). For future computational pathology and medical imaging studies, we recommend: (1) mandatory patient/biopsy-level `GroupKFold` splitting; (2) foreground-only or zero-penalty Dice reporting on sparse-lesion datasets; (3) 95% confidence intervals on every single-run comparison; (4) validation on diverse multi-organ benchmarks such as PanNuke; and (5) systematic evaluation of multi-task balancing hyperparameters before clinical translation.

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